exercises_SSD

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Settings

Required libraries:

```
library(devtools)   # Installing package
library(lme4)   # Multilevel model
library(bain)   # Bayes factor
```

Installing the Package

```
# For older version of devtools
install_github("ulrichlosener/BayesSSD")

# For latest version of devtools
pak::pak("ulrichlosener/BayesSSD")

library(BayesSSD)
```

Longitudinal Interventions

Survival function

The sample size determination is designed as two-step process. It is advised to first determine the survival function, which defines the expected drop-out pattern of participants. This can be done using the Shiny app Attrition.

Note: Focus only on non-intermittent missingness, that is, drop-out/attrition!

Additional Exercises 1. First year students

A health psychology team investigates how first-year university students adapt to academic stress during their transition into higher education. Participants are recruited during orientation week at a large university and are introduced to a smartphone-based study that tracks stress, mood, and social context in real time. The protocol involves responding to brief surveys five times per day over a four-week period, with reminders delivered at semi-random intervals. While the onboarding sessions emphasize the importance of capturing experiences in the moment, students only fully encounter the pace and intrusiveness of the prompts once their regular class schedules begin and daily routines take shape. The students have their orientation week in week 1 of the study, and their first preliminary exams in the week 4.

2. Employees in a large company

A research team in organizational psychology studies how employees adapt to a new digital workflow system introduced within a large company. Participants complete short surveys twice a week over a 12-week period, reporting on productivity, frustration, and perceived efficiency. In the initial phase, the novelty of the system and accompanying training sessions keep employees attentive to both the tool and the study. As the rollout

stabilizes and the system becomes part of everyday work, enthusiasm for reflecting on the process appears to wane, especially before the organization begins actively preparing summary evaluations toward the end of the study period. The summary evaluations are highly anticipated by many participants, who are interested in the personalized results.

3. Adolescents

A developmental psychology study follows adolescents over the course of a school semester to examine changes in peer relationships and well-being. Students are asked to complete brief weekly surveys during recess, with teachers providing occasional reminders and encouragement. At the start of the semester, students are still adjusting to new schedules and classmates, and participation routines are not yet firmly established. Midway through the term, survey completion becomes more integrated into classroom routines, before the final weeks bring increasing academic pressure, exams, and end-of-term activities that reshape students' priorities and availability.

4. Sleep lab

A sleep research lab conducts a longitudinal trial to evaluate the effectiveness of a behavioral intervention aimed at improving sleep quality among adults with mild insomnia. The study spans six months and includes monthly online check-ins assessing sleep patterns, daytime functioning, and adherence to recommended routines. Throughout most of the study, participants receive automated summaries of their sleep data and occasional encouragement messages. The final phase of the study involves a comprehensive in-person assessment at the lab, including interviews and physiological measurements scheduled at flexible but limited time slots.

5. Lifestyle program

A public health research team investigates changes in physical activity among adults participating in a 10-week lifestyle program designed to promote regular exercise. Participants are equipped with wearable fitness trackers and are asked to complete weekly reflections on their activity levels, motivation, and perceived barriers. The program includes optional group challenges and periodic feedback summaries that highlight individual progress relative to personal goals. Across the study period, participants show diverse trajectories in how their activity patterns evolve, shaped by factors such as work schedules, prior fitness levels, and everyday constraints.

Sample Size Determination

The second step is to include the attrition function and relevant information for the sample size (for the effect sizes and variance-covariance components, you may stick to the defaults for simplicity). Note that omega refers to the proportion of individuals dropping out and gamma to the timing of drop-out.

```r
SSD_longit(attrition = "weibull", # survival function to model attrition
           params = c(0.5, 1),    # omega and gamma
           m = 1000,              # number of simulated datasets per iteration
           seed = 123)
```

```
  |                                                                      |

Final sample size
=================
Hypotheses:
    H1: a<b<c
    Hc: not H1
Using  a total of N = 104 Participants
P (BF_1c > 5 | H1) =  0.799
```

Cluster Randomised Trials

For two-arm Cluster Randomised Trial it is important to define beforehand the hypotheses of study.

- Pair 1: Null vs. alternative

$H_0 : \mu_C = \mu_T$

$H_1 : \mu_C < \mu_T$

- Pair 2: Informative hypotheses

$H_1 : \mu_C < \mu_T$

$H_2 : \mu_C > \mu_T$

```
# Pair 1
SSD_crt_null(eff_size = 0.2,          # Effect size
             n1 = 20,                 # Cluster size
             ndatasets = 100,         # Number of datasets to generate
             rho = 0.1,               # Intraclass correlation coefficient
             BF_thresh1 = 3,          # Bayes factor threshold if H1 is true
             BF_thresh0 = 3,          # Bayes factor threshold if H0 is true
             eta1 = 0.7,              # Desired power if H1 is true
             eta0 = 0.7,              # Desired power if H0 is true
             fixed = "n1",            # The sample size that is fixed
             b_fract = 2,             # b fraction to run the sensitivity analysis
             max = 200,               # Maximum sample size that can be reached
             batch_size = 50)         # Batch size for fitting the multilevel model
```

```
Final sample size
=================
Hypotheses:
    H0: Intervention=Control
    H1: Intervention>Control
**********************************************************************
     b        n1       n2       P(BF.01 > 3 | H0) >  0.7
[1,] "  1.00" " 20.00" "104.00" "  0.94"
[2,] "  2.00" " 20.00" " 84.00" "  0.92"
     P(BF.10 > 3 | H1) >  0.7
[1,] "  0.76"
[2,] "  0.73"
**********************************************************************
n1: Cluster sizes
n2: Number of clusters
```

The output includes a table. The second column contains the number of clusters whilst the third column contains the cluster size. The fourth and fifth columns contain the empirical power when the null and alternative hypotheses are true, respectively. The output includes a sensitivity analysis using different values for fraction b (first column).

```
# Pair 2
SSD_crt_inform(eff_size = 0.5,          # Effect size
               n1 = 20,                 # Cluster size
               ndatasets = 100,         # Number of datasets to generate
               rho = 0.1,               # Intraclass correlation coefficient
               BF_thresh = 3,           # Bayes factor threshold
               eta = 0.8,               # Desired power
```

```
                fixed = "n1")            # The sample size that is fixed
```

```
Final sample size
=================
Hypotheses:
    H1: Intervention1 > Intervention2
    H2: Intervention1 < Intervention2
Using cluster size =  20  and number of clusters =  8
P (BF.12 >  3  | H1) =  0.97
```

The output include the cluster size and number, the number of clusters, and the empirical power.

In addition to the package, the Shiny app SSD for CRT with Bayes factor can be used to determine the sample size for a set of common values of effect size, ICC, number of clusters, and cluster sizes.

Additional exercises

Situation 1 A

A research teams is designing a study to evaluate the effectiveness of a smoking prevention programme for children combining the current intervention at school with an additional intervention at home. The researchers decided to assign randomly schools to one of the intervention conditions

- School-Only (control): Prevention curriculum delivered during school hours.
- School + Home: The standard curriculum plus an integrated home-based family component.

According to previous literature about similar interventions, the researchers expect a Cohen's d of 0.2 and an ICC of 0.1. A total of 60 schools have agreed to participate in the study. The researchers decide to aim for a Bayes factor of at least 5 to provide an evidence they consider substantial. Given the values mentioned, determine the required number of students per school necessary.

Situation 1 B

A research teams is designing a study to evaluate which smoking prevention programme for children is better. The researchers decided to assign randomly schools to one of the intervention conditions

- School-Only: Prevention curriculum delivered during school hours.
- School + Home: The standard curriculum plus an integrated home-based family component.

According to previous literature about similar interventions, the researchers expect a Cohen's d of 0.5 and an ICC of 0.025. A total of 60 schools have agreed to participate in the study. The researchers decide to aim for a Bayes factor of at least 5 to provide an evidence they consider substantial. Given the values mentioned, determine the required number of students per school necessary.

Situation 2 A

A group of researcher have developed a new programme called FUNBALL to prevent injuries among football players. In the study, teams assigned to the intervention condition will complete FUNBALL sessions after their usual warm-up, while the control condition will maintain their usual training routine.

Based on training data and previous literature, the research group expects an ICC of 0.025 and an effect size of 0.8. Preliminary recruitment efforts indicate that 40 players per team are willing to participate. Using the Bayesian hypothesis testing, the researchers seek a Bayes factor of at least 2. Given this information, determine the total number of teams that must be recruited for the study.

Situation 2 B

A group of researcher have developed a new programme called FUNBALL to prevent injuries among football players. To evaluate its efficacy, the researchers have designed a cluster randomised trial comparing the

FUNBALL program to a standard training control. In the study, teams assigned to the FUNBALL condition will complete FUNBALL sessions, while the other condition will complete the well stablished FIFA 11+ programme.

Based on training data and previous literature, the research group expects an ICC of 0.05 and an effect size of 0.5. Preliminary recruitment efforts indicate that 5 players per team are willing to participate. Using the Bayesian hypothesis testing, the researchers seek a Bayes factor of at least 5. Given this information, determine the total number of teams that must be recruited for the study.

Situation 3 A

A new programme involving Health visitor training in the clinical assessment of postnatal women is being evaluated. The group of researchers compares the effectivity of the new programme with the standard postnatal care. In the cluster randomised trial, the researchers will allocate randomly the health centres into the intervention and the control conditions.

Based on previous clinical trials in maternal mental health, the researchers expect an ICC of 0.1 and an effect size (Cohen's d) of 0.8. Due to the average number of mothers at the centres, the cluster size is fixed at 40. The desired Bayes factor is at least 3. Considering this information, determine the number of clusters required.

Situation 3 B

A clinical research team is investigating which psychological framework is more effective for treating postnatal depression: a Cognitive Behavioural Approach (CBA) or a Person-Centred Approach (PCA). In the cluster randomized trial, the health centres are assigned to one of the two approaches.

Based on previous clinical trials in maternal mental health, the researchers expect an ICC of 0.1 and an effect size (Cohen's d) of 0.8. Due to the average number of mothers at the centres, the cluster size is fixed at 30. The desired Bayes factor is at least 3. Considering this information, determine the number of clusters required.

Based on:

Efficacy of a new injury prevention programme (FUNBALL) in young male football (soccer) players: a cluster-randomised controlled trial - PubMed

Psychological interventions for postnatal depression: cluster randomised trial and economic evaluation. The PONDER trial | NIHR Journals Library

References

Barragan I., C. N. , & Moerbeek, M. (2025). Method for Sample Size Determination for Cluster-Randomized Trials Using the Bayes Factor. *Journal of Educational and Behavioral Statistics*, *0*(0). https://journals.sagepub.com/doi/10.3102/10769986251362404

Barragan Ibañez CN, Lösener U, Moeteke N, Moerbeek M (2026) Design and analysis of behavioral intervention studies: A Bayesian approach. PLoS One 21(2): e0342163. https://doi.org/10.1371/journal.pone.0342163